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BIG DATA ANALYTICS

Assignment 2

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Assignment 2 Data Source Selection & Ingestion Pipeline

Repo link: <https://github.com/hassaanawan48/NYC-Taxi-Ingestion-Pipeline>

Dataset Justification

Selected dataset: NYC Yellow Taxi Trip Records (Jan 2024)

URL: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

File: yellow_tripdata_2024-01.parquet

Dataset Description

2,964,624 raw trip records. 19 columns: datetime, numeric, categorical, and location IDs.

Warehouse Suitability

1. Volume exceeds 500k rows, justifies HDFS/Spark.
2. Each row is a trip, natural fact table.
3. Clear dimensions: time, location, payment, vendor.
4. Raw data contains missing values, outliers, negative fares, ideal for ETL in A3.

Business Questions (5)

1. How do trip volume and average fare vary by hour and day of week?
2. What is the correlation between trip distance and fare amount?
3. How does payment type preference change by time and location?
4. How does average tip differ by payment type, distance, or vendor?
5. Which pickup/dropoff location pairs have highest volume and fare?

Size verification:

```
import pandas as pd
df = pd.read_parquet("yellow_tripdata_2024-01.parquet")
print(f"Rows: {len(df):,}")
print(f"Columns ({len(df.columns)}): {list(df.columns)}")
```

```
Rows: 2,964,624
Columns (19): ['VendorID', 'tpep_pickup_datetime', 'tpep_dropoff_datetime', 'passenger_count', 'trip_distance', 'RatecodeID', 'store_and_fwd_flag', 'PULocationID', 'DOLocationID', 'payment_type', 'fare_amount', 'extra', 'mta_tax', 'tip_amount', 'tolls_amount', 'improvement_surcharge', 'total_amount', 'congestion_surcharge', 'Airport_fee']
```